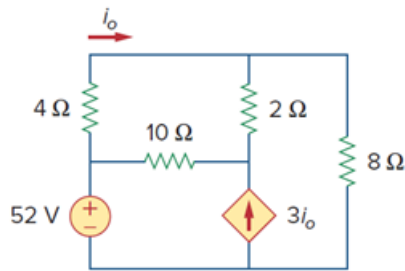


Problem

Use mesh analysis to find the current i_o in the circuit of Fig. 3.95.

Figure 3.95:



Step-by-step solution

Step 1 of 5

Refer to Figure 3.95 in the textbook.

Draw the circuit diagram with mesh currents.

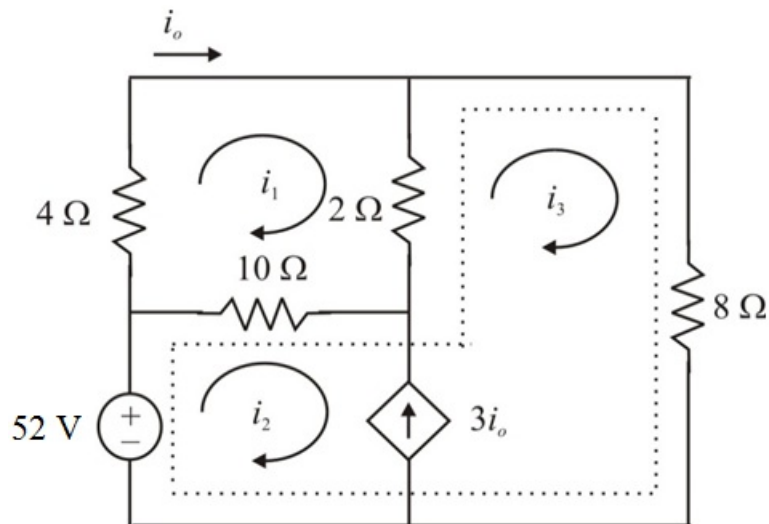


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law to Loop 1.

$$4i_1 + 2(i_1 - i_3) + 10(i_1 - i_2) = 0$$

$$4i_1 + 2i_1 - 2i_3 + 10i_1 - 10i_2 = 0$$

$$16i_1 - 10i_2 - 2i_3 = 0$$

$$8i_1 - 5i_2 - i_3 = 0 \quad \dots\dots (1)$$

Apply Kirchhoff's voltage law to Loops 2 and 3 together.

$$-52 + 10(i_2 - i_1) + 2(i_3 - i_1) + 8i_3 = 0$$

$$-52 + 10i_2 - 10i_1 + 2i_3 - 2i_1 + 8i_3 = 0$$

$$-12i_1 + 10i_2 + 10i_3 = 52 \quad \dots\dots (2)$$

Step 3 of 5

From Figure 1, the output current is,

$$i_o = i_1$$

From the Figure 1, the current through the dependent current source is,

$$3i_o = i_3 - i_2$$

Substitute i_1 for i_o .

$$3i_1 = i_3 - i_2$$

$$3i_1 + i_2 - i_3 = 0$$

$$i_3 = 3i_1 + i_2 \dots\dots (3)$$

Step 4 of 5

Substitute $3i_1 + i_2$ for i_3 in equation (1).

$$8i_1 - 5i_2 - (3i_1 + i_2) = 0$$

$$5i_1 - 6i_2 = 0$$

$$6i_2 = 5i_1$$

$$i_2 = \frac{5}{6}i_1 \dots\dots (4)$$

Substitute $3i_1 + i_2$ for i_3 in equation (2).

$$-12i_1 + 10i_2 + 10(3i_1 + i_2) = 52$$

$$18i_1 + 20i_2 = 52 \dots\dots (5)$$

Step 5 of 5

Substitute $\frac{5}{6}i_1$ for i_2 in equation (5).

$$18i_1 + 20\left(\frac{5}{6}i_1\right) = 52$$

$$18i_1 + 10\left(\frac{5}{3}i_1\right) = 52$$

$$54i_1 + 50i_1 = 52(3)$$

$$104i_1 = 156$$

$$i_1 = 1.5 \text{ A}$$

The value of output current i_o is,

$$i_o = i_1$$

$$= 1.5 \text{ A}$$

Therefore, the value of output current, i_o is 1.5 A.